# Goldsmiths - University of London

## INTEGRATED DEGREE IN PSYCHOLOGY

### Foundation Year 2025-26

## PS50008A: RESEARCH METHODS AND EXPERIMENTAL DESIGN

# PRACTICE MCQ TEST

**STUDENT ID:** ________________________________ **DATE:** ______________

FOR EACH QUESTION CHOOSE ONE ANSWER AND CIRCLE IT

This is a multiple-choice assessment. You should attempt ALL 36 questions.

**You are reminded that:**

- There are 5 investigations, each with 6 questions, plus 6 general questions.
- Read each question and set of answers carefully before making your choice.
- For **EACH** question choose **ONE** of the answer options and put a **CIRCLE** around the **LETTER** that corresponds to your choice.
- If you wish to change your answer, then clearly strike through and circle an alternative.

**INVESTIGATION 1**

A researcher investigated whether sleep deprivation affects reaction time. 30 participants were randomly allocated to either a sleep-deprived condition (4 hours sleep) or a control condition (8 hours sleep). The next morning, all participants completed a computer task measuring reaction time in milliseconds. The researcher found that sleep-deprived participants had significantly slower reaction times ($p<.05$).

**1.** What kind of design did the researcher use?

(a) a within participants (repeated measures) design
(b) a correlational design
(c) a between participants (independent participants) design
(d) a matched pairs design

**2.** What was the independent variable?

(a) there was no independent variable
(b) the amount of sleep (4 hours/8 hours)
(c) the reaction time
(d) the participants

**3.** What was the dependent variable?

(a) there was no dependent variable
(b) the amount of sleep (4 hours/8 hours)
(c) the reaction time in milliseconds
(d) the computer task

**4.** What was the researcher's two-tailed hypothesis?

(a) that sleep-deprived participants would have significantly slower reaction times
(b) that sleep-deprived participants would have significantly faster reaction times
(c) that there would be a significant difference in reaction times between the two groups
(d) that there would be a significant correlation between sleep and reaction time

**5.** What statistical test should the researcher use?

(a) a Spearman's rho test
(b) a paired-samples t-test
(c) an independent-samples t-test
(d) a Pearson's r test

**6.** Did the results support the researcher's hypothesis?

(a) yes
(b) no
(c) the results are unclear
(d) more data is needed

**INVESTIGATION 2**

A researcher wanted to know whether anxiety levels are related to exam performance. 40 students completed an anxiety questionnaire (scored 0-50, higher scores indicate more anxiety) one week before their exam. After the exam, the researcher recorded each student's percentage score. The analysis revealed a significant negative correlation ($p<.01$).

**7.** What kind of design did the researcher use?

(a) a within participants (repeated measures) design
(b) a correlational design
(c) a between participants (independent participants) design
(d) a matched pairs design

**8.** What was the independent variable?

(a) the anxiety scores
(b) the exam scores
(c) the students
(d) there was no independent variable

**9.** What was the researcher's one-tailed hypothesis?

(a) that there would be a significant positive correlation between anxiety and exam scores
(b) that there would be a significant negative correlation between anxiety and exam scores
(c) that there would be a significant difference between anxiety and exam scores
(d) that students with high anxiety would score exactly 50 on the questionnaire

**10.** What statistical test should the researcher use?

(a) an independent-samples t-test
(b) a paired-samples t-test
(c) a Pearson's r test
(d) no test was necessary

**11.** What does a "significant negative correlation" mean in this context?

(a) higher anxiety is associated with higher exam scores
(b) higher anxiety is associated with lower exam scores
(c) there is no relationship between anxiety and exam scores
(d) anxiety causes poor exam performance

**12.** How often would this pattern of results have occurred by chance?

(a) more than 5 times in a hundred
(b) less than or equal to 5 times in a hundred
(c) less than or equal to 1 time in a hundred
(d) none of the above

**INVESTIGATION 3**

A researcher investigated whether background music affects concentration. 24 participants each completed two proofreading tasks (finding spelling errors in a passage): once with classical music playing and once in silence. The order was counterbalanced. The number of errors detected was recorded. Results showed participants found significantly more errors in the silence condition ($p<.05$).

**13.** What kind of design did the researcher use?

(a) a between participants (independent participants) design
(b) a correlational design
(c) a within participants (repeated measures) design
(d) a matched pairs design

**14.** What was the independent variable?

(a) there was no independent variable
(b) the background condition (music/silence)
(c) the number of spelling errors detected
(d) the participants

**15.** What was the dependent variable?

(a) the background condition (music/silence)
(b) the number of spelling errors detected
(c) the participants
(d) there was no dependent variable

**16.** Why did the researcher counterbalance the order?

(a) to ensure equal numbers in each condition
(b) to control for practice or fatigue effects
(c) to increase the sample size
(d) to make the study correlational

**17.** What statistical test should the researcher use?

(a) a Pearson's r test
(b) an independent-samples t-test
(c) a paired-samples t-test
(d) a Spearman's rho test

**18.** Did the results support the researcher's hypothesis?

(a) yes
(b) no
(c) the results are unclear
(d) more data is needed

**INVESTIGATION 4**

A researcher investigated whether phone notifications affect study performance. 50 participants were randomly assigned to one of two groups. One group studied a chapter with their phone notifications turned on, while the other group studied the same chapter with notifications turned off. All participants then completed a comprehension quiz. The researcher found no significant difference between the groups ($p>.05$).

**19.** What kind of design did the researcher use?

(a) a within participants (repeated measures) design
(b) a correlational design
(c) a between participants (independent participants) design
(d) a matched pairs design

**20.** What was the independent variable?

(a) there was no independent variable
(b) the phone notification condition (on/off)
(c) the comprehension quiz score
(d) the chapter being studied

**21.** What was the dependent variable?

(a) the phone notification condition (on/off)
(b) the comprehension quiz score
(c) there was no dependent variable
(d) the participants

**22.** What was the researcher's two-tailed hypothesis?

(a) that participants with notifications on would score significantly higher
(b) that participants with notifications off would score significantly higher
(c) that there would be a significant difference in quiz scores between the two groups
(d) that there would be a significant correlation between notifications and quiz scores

**23.** What statistical test should the researcher use?

(a) a paired-samples t-test
(b) an independent-samples t-test
(c) a Pearson's r test
(d) a Spearman's rho test

**24.** Did the results support the researcher's hypothesis?

(a) yes
(b) no
(c) the results are unclear
(d) more data is needed

## INVESTIGATION 5

A researcher investigated whether exercise is related to self-reported mood. 60 participants recorded how many hours of exercise they did in a typical week and also completed a mood questionnaire (scored 1-100, higher scores indicate better mood). The researcher found a significant positive correlation between weekly exercise hours and mood scores ($p<.05$).

**25.** What kind of design did the researcher use?

(a) a within participants (repeated measures) design
(b) a correlational design
(c) a between participants (independent participants) design
(d) an experimental design

**26.** What was the independent variable?

(a) the hours of exercise
(b) the mood scores
(c) the participants
(d) there was no independent variable

**27.** What was the researcher's one-tailed hypothesis?

(a) that there would be a significant negative correlation between exercise and mood
(b) that there would be a significant positive correlation between exercise and mood
(c) that there would be a significant difference between exercise and mood
(d) that people who exercise would have exactly the same mood

**28.** What statistical test should the researcher use?

(a) an independent-samples t-test
(b) a paired-samples t-test
(c) a Pearson's r test
(d) no test was necessary

**29.** What does a "significant positive correlation" mean in this context?

(a) more exercise is associated with better mood
(b) more exercise is associated with worse mood
(c) there is no relationship between exercise and mood
(d) exercise causes better mood

**30.** Can the researcher conclude that exercise causes better mood?

(a) yes, because the correlation was significant
(b) yes, because the p-value was less than .05
(c) no, because correlation does not imply causation
(d) no, because the sample size was too small

## GENERAL QUESTIONS

**31.** A researcher wants to study "aggression." Which of the following is an operationalisation of aggression?

(a) the feeling of being angry at someone
(b) the number of times a child hits another child during playtime
(c) a negative emotional state
(d) the opposite of kindness

**32.** What does it mean if a study is "replicable"?

(a) it used a large sample size
(b) the same results can be obtained when the study is repeated
(c) the results were statistically significant
(d) the study was published in a journal

**33.** A confounding variable is:

(a) the variable the researcher manipulates
(b) the variable the researcher measures
(c) an uncontrolled variable that could affect the results
(d) the same as the dependent variable

**34.** A two-tailed hypothesis:

(a) predicts the direction of the effect
(b) predicts there will be a difference but not the direction
(c) predicts no difference will be found
(d) is the same as the null hypothesis

**35.** A p-value of $p<.05$ means:

(a) the results are definitely true
(b) there is less than a 5% probability the results occurred by chance
(c) 5% of participants dropped out
(d) the effect size is large

**36.** The group of people a researcher actually studies is called the:

(a) population
(b) sample
(c) control group
(d) experimental group

**END OF PRACTICE TEST**